System Architecture – ML Visualization Tool

FRONTEND LAYER (React 18 + Vite)

7-Step Navigation
(Step 0–7) + Progress Bar

Domain Pill Bar
(20 Medical Specialties)

Step 1:
Clinical Context Page

Step 2:
Data Upload & Column Paper

Privacy Notice (Visible on All Screens)

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REST API – JSON over HTTPS

```
GET /domains
GET /context?domain=...
POST /data/upload
POST /data/default
POST /data/map-columns
```

BACKEND LAYER (FastAPI - OpenAPI Enabled)

REST API Layer
(Stateless)

Domain Controller

Data Manager

Validation Service
(file size / row constraints)

Session Manager

Preprocessing Service
(planned)

Model Service
(planned)

Evaluation Service
(planned)

Explainability Service
(planned)

Certificate Generator
(planned)

Temporary In-Memory Session Storage Only
No Persistent Database
(GDPR-Compliant Educational Tool)

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ML Pipeline Execution
(Preprocess → Train → Evaluate → Explain)

→ **Metrics JSON** → **Confusion Matrix Data** → **Feature Importance Data**

MACHINE LEARNING LAYER (scikit-learn)

Preprocessing

- Train/Test Split
- Missing Values (Median, Mode, Remove)
- Normalization (Z-score, Min-Max)
- SMOTE (optional)

Models

- KNN
- SVM
- Decision Tree
- Random Forest
- Logistic Regression
- Naive Bayes

Evaluation

- Accuracy
- Sensitivity
- Specificity
- Confusion Matrix

Explainability

- Feature Importance
- Single Patient Explanation